

602 Project

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2024-02-13

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## New names:  
## * '' -> '...37'
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Introduction

Cancer is one of the most complicated and rare, yet prominent diseases across our population. The unique differences between each type of cancer makes oncology akin to throwing darts in the dark. Precision oncology aims to change this. Precision oncology is the practice of personalizing cancer treatment based on a person's genetic makeup. Medicine has historically relied on qualitative data to determine the solution to cancer, however this is changing with the introduction of Next Generation Sequencing. Next Generation Sequencing allows us to sequence patient's whose root cause was not identified by diagnostics. Years of sequencing led to the development of robust databases of genes and their effect on each person. The culmination of this data lets us draw conclusions across populations.

The Genome Aggregation Database (gnomAD) is the international result of testing across many different populations. The aims of this database is to represent a "given person". Any major genetic outliers are not included. GnomAD tracks individual mutations across populations checking for every little change in amino acids (the building blocks of genes). These amino acid changes are known as mutations. There are several different kinds of mutations which can range from doing nothing (silent mutations) to disrupting cell repair (oncogenes). Every mutation gnomAD has seen is recorded for each of their populations in a Variant Calling File. This file is a CSV that records the genetic location of the mutation, as well as how prevalent it is in each population. With this, we are able to see which mutations are more prevalent in one population than another.

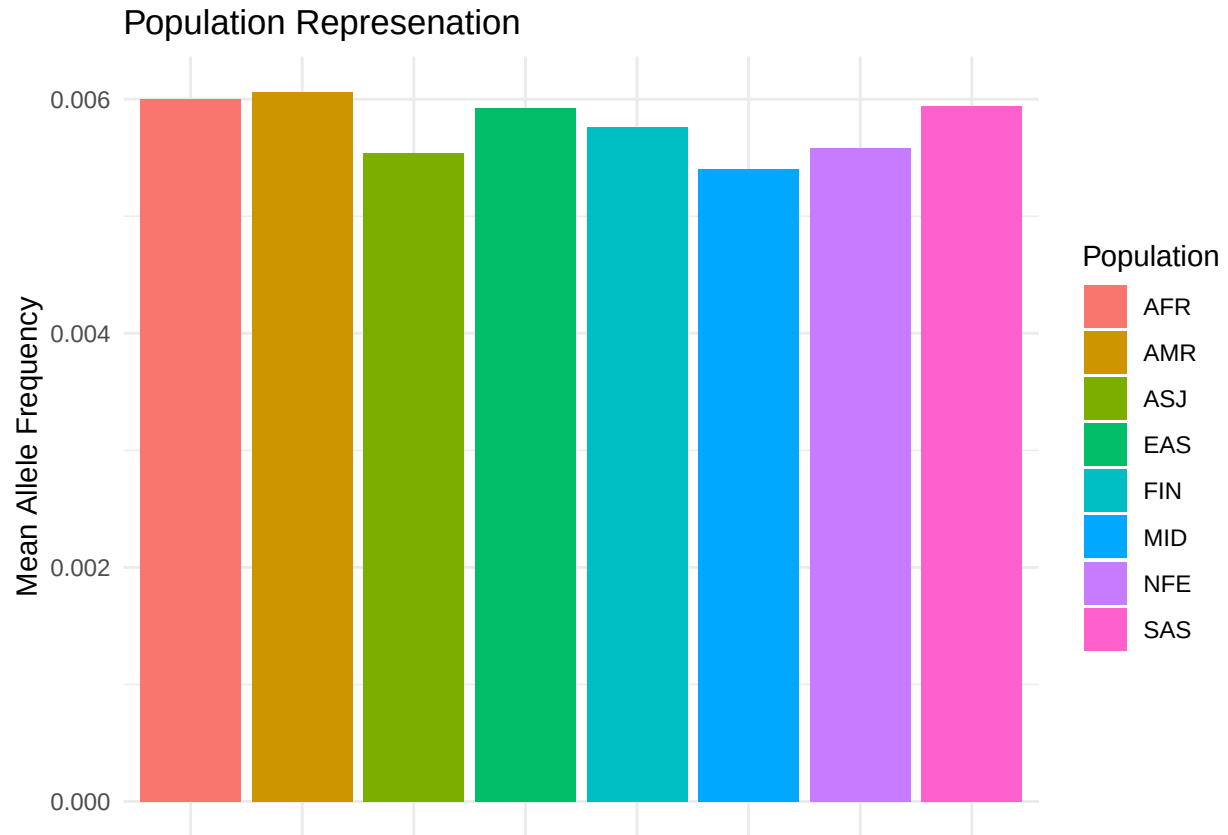
ClinVar is another genetic database that functions alongside gnomAD. The purpose of ClinVar is to record clinical histories of individual mutations. For example, if a BRCA1 gene was seen to be mutated in a specific way, that clinic would report their findings on if that mutation is associated with the patient's cancer.

Historically gnomAD has been used by clinical geneticists who query a gene they found in their patient to learn more about it. It's used for specific documentation of rare mutations and their effects. ClinVar historically is a research database. It is used to compare results from different labs to determine the meaning of a mutation, as well as if their diagnostics are in line with the rest of the community.

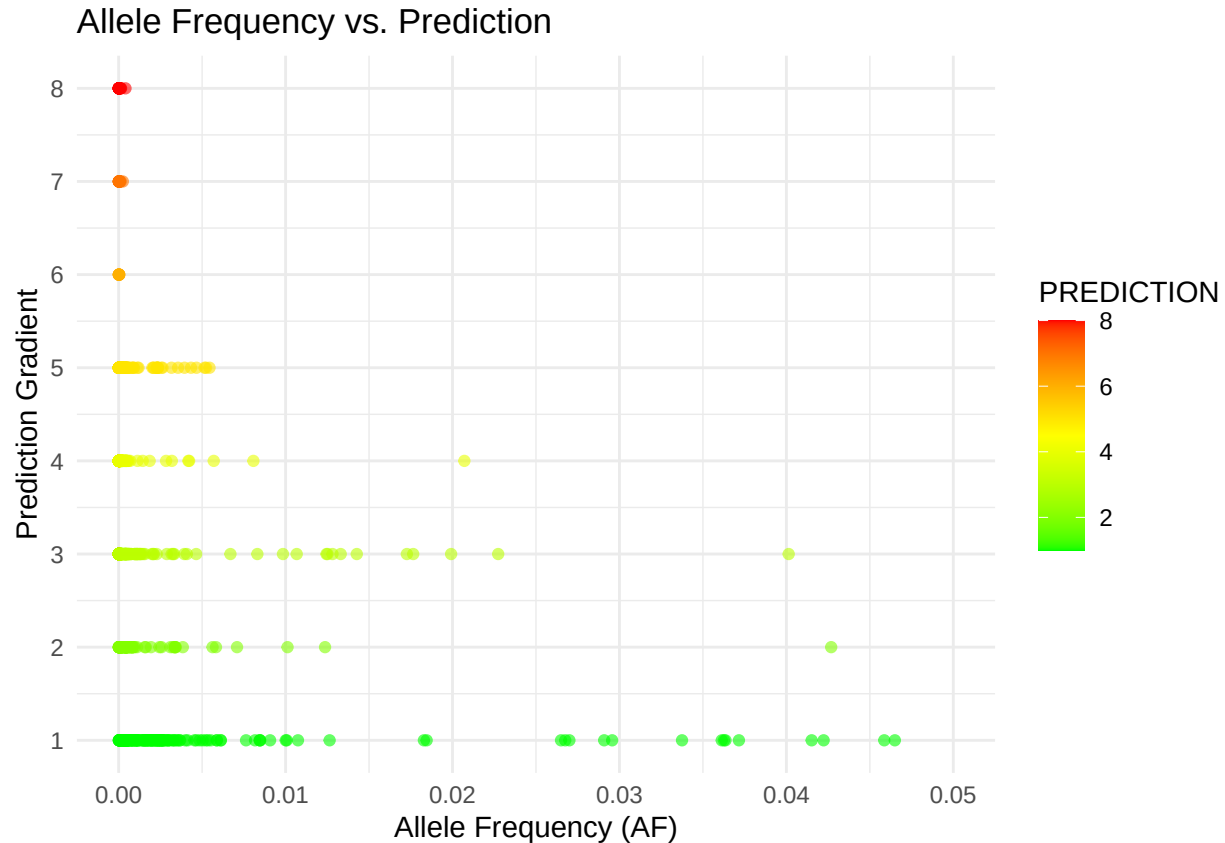
Our resulting data is the ClinVar gene and prediction intersected with the gnomAD population data. This allows us to perform analysis on whether population data is linked to the pathogenicity of the mutation. Pathogenicity is the measurement of how likely a mutation is to be a driver of cancer. If it is not likely, the mutation is said to be benign. Population data from gnomAD tracks Allele Frequency, which is how often we see this mutation in the population. A rule from the American Council of Medical Geneticist is that if the frequency of a mutation is greater than 5%, it can be classified as benign. This is because if a mutation is that common it is extremely unlikely to be the driver of cancer. If it would be, then 5% of the population would have that type of cancer (or predisposed to).

Motivation

The motivation of this analysis is to test the validity of population genetics. As precision medicine grows more prominent, the best tests and diagnostics remain out of reach for the less fortunate. Population genetics is a way to bring precision medicine to those who are unable to receive sequencing. If there is a correlation between Allele frequency and the Clinvar prediction, populations and individuals can understand their genetic predisposition based on gene.



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## Warning: Removed 91 rows containing missing values ('geom_point()').
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Data Visualization Graphs

These two graphs represent the dataset in different ways. The bar graph shows that the mean allele frequency for each population. This is to show that although some populations have more samples in gnomAD than others, all populations have some mutations that are common in theirs but not other populations.

The scatter plot shows the rough distribution of Allele frequency versus Prediction. This graph shows that the trend is higher allele frequency tends to create a more distinct benign or likely benign prediction (1 and 2 respectively). Lower allele frequency produces more pathogenic variants, however they also produce many benign variants.

Guiding Questions Is allele frequency an accurate measure of pathogenicity? How well does Allele frequency correlate to Benign predictions? How about Pathogenic predictions? Are these predictions in line with the Automatic Benign rule from the ACMG?

Determining if Allele frequency is an accurate measure for benign is important because it allows us to prescribe impact to mutations without a complete analysis. For example, if a cancer patient was found to have a BRCA1 mutation that the geneticist is not aware of. It is also important to check our ClinVar predictions against the ACMG automatic benign rule, as the publications do not necessarily follow this rule.

Model Building

Hypotheses

Null Hypothesis: There is no association between ClinVar's prediction and Allele Frequency. Alternate Hypothesis: As Allele Frequency becomes larger, ClinVar's prediction will shift towards benign. ClinVar's prediction will follow the Automatic Benign rule.

Model Fitting

$$Y_i = \beta_0 + \beta_1 X_i + e_i$$

Where $\hat{\beta}_0$ is the intercept and $\hat{\beta}_1$ is the slope.

To create our linear regression to make a formula of our own, we first must convert the Pathogenicity predictors into numerical categorical variables. The conversion was done by attributing 1 to “Benign” and 8 to “Pathogenic”. Predictors such as “Pathogenic/Likely Pathogenic” and “Benign/Likely Benign” are therefore attributed to 7 and 2 respectively. We are then able to investigate the regression where Y is the Prediction, and X is allele frequency.

Model Evaluation

Significance of the Coefficient Estimates. If our slope is to be equal to 0, our null hypothesis evaluation is correct.

$$H_0, \beta_1 = 0$$

However, if our slope is not equal to 0, it indicates correlation between Allele Frequency and Prediction.

$$H_a, \beta_1 \neq 0$$

In this case; $\hat{\beta}_1 = -0.088272$. This means as Allele frequency drops, our prediction severity increases. This is inline with our hypothesis and the Automatic Benign rule. The $\hat{\beta}_0 = 0.229275$ does not have a meaning as Prediction is a categorical variable ranging from 1-4, with 0 representing “others”.

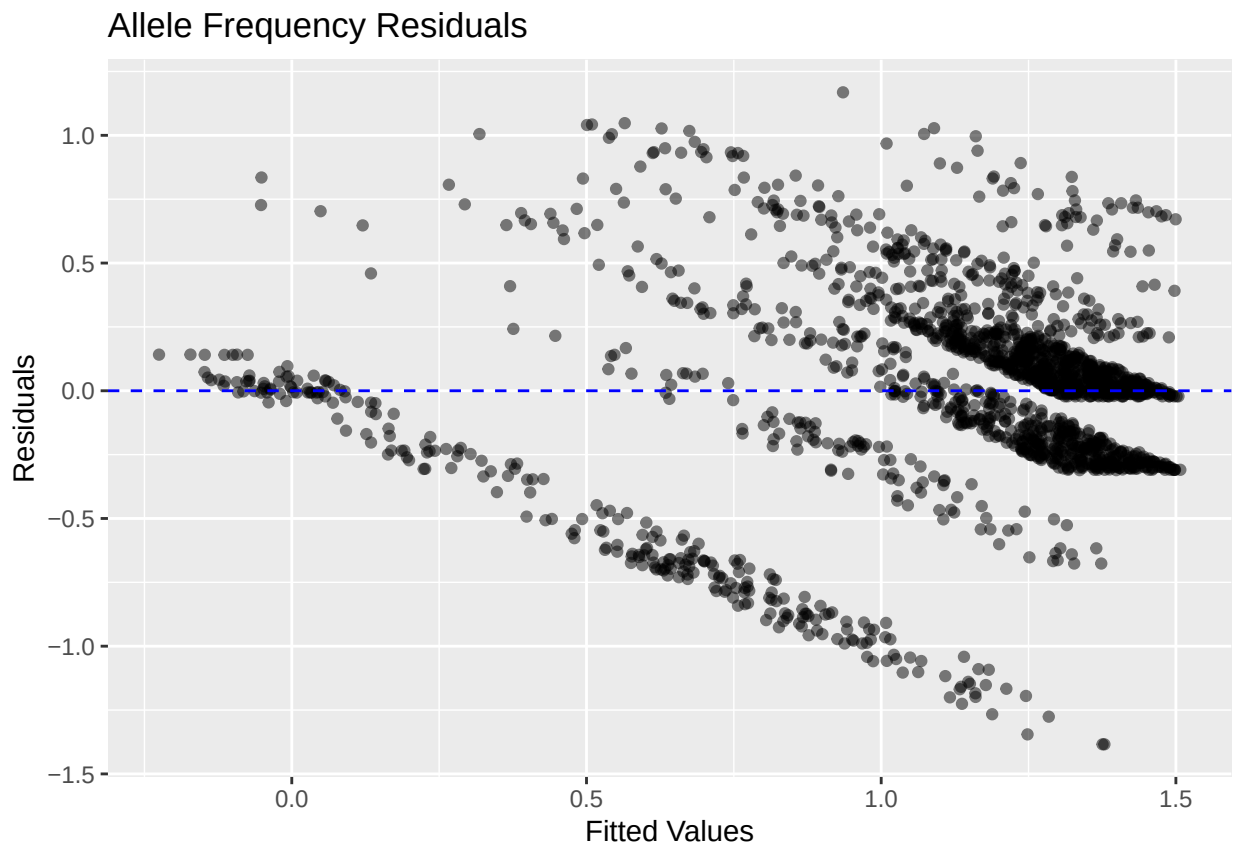
We can now reclassify our model as:

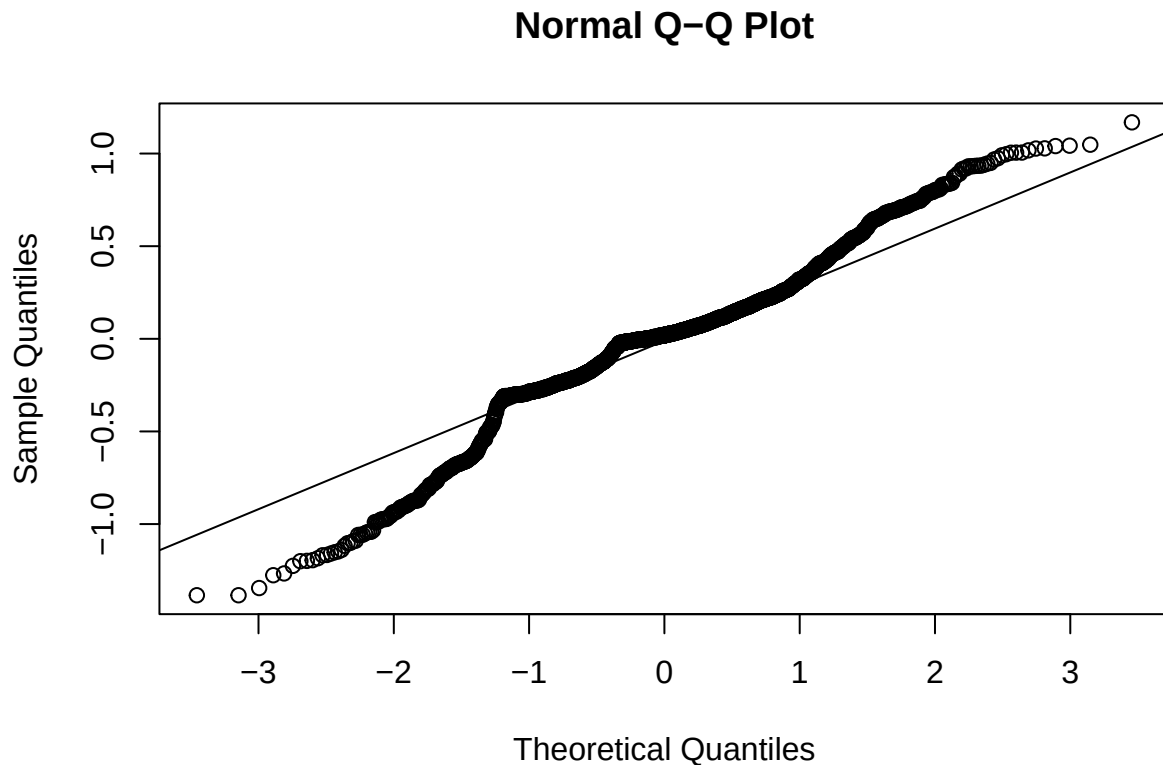
$$Prediction_i = 0.229275 + -0.088272 \log(AF_i) + e_i$$

This formula the predicted value of ClinVar’s prediction based on a log Allele Frequency. The negative slope value indicates there is a negative relationship between Allele Frequency and Prediction. This correlates with our estimate, that population genetics (5% rule) will be in line with Clinvar’s clinical findings.

Our model produces an R squared value of 0.314, which means 31.4% of the variance can be explained by the variable in the model. This means a noticeable portion of the data cannot be explained by the model. We prescribe this to the “middle” values that exist in the dataset. These values are of uncertain significance or have conflicting interpretations. This means that while it says uncertain, there are multiple cases where that mutation was pathogenic.

Residual Analysis





```
##
##  Shapiro-Wilk normality test
##
## data:  residuals
## W = 0.96648, p-value < 2.2e-16
```

The qqplot shows linear distributed values, except it varies at the tails. This means that there are some outliers within the Allele Frequency that are extremely small. Even with outlier removal via quartiles, AF values can vary drastically from gene, to population, to mutation. The equal variance plot is based off of categorical variables and therefore comes in bands. This banding can be visually improved by jittering, but still shows a drawback with categorical simple linear regressing.

Conclusion

Final Model

The model shows that there is a negative correlation between prediction, and Allele Frequency. Prediction is scored on a 1-8 system, where 1 is the most certain benign claims, and 8 is the most certain pathogenic claim. This is in line with the Automatic Benign rule, which states that if Allele frequency is too high in a population the mutation can be classified as benign. With this; outliers in this dataset proved difficult and introduced error into the analysis.

We can reject our Null hypothesis in favor of our alternate, considering the p value was extremely small at (2^{-16}), meaning there is a strong association. We also found that association to be negative from the slope (-0.088272).

we can then say the mutation predictions from ClinVar are accurately assessing mutations, and are in line with the Automatic Benign rule. It's important to note ClinVar and gnomAD are not associated. ClinVar simply has some information that overlaps with variants present in the monstrous amount of gnomAD data. We tested whether Allele frequency can accurately determine when a variant should be determined as "benign". The 5% rule is standard that we placed our model against. What we found is that Allele frequency was a statistically significant indicator when producing claims whether a variant is benign. This means that the frequency of an allele might be the reason why ClinVar's clinical studies predicted them as benign.

Prediction with Allele Frequency of 0.05

```
## [1] 0.8999113 2.9828751
```

If the Allele Frequency is high (0.05), we are then expecting to see a ClinVar prediction of Benign or Likely Benign, which follows the Automatic Benign rule. Using our formula, if we pass in 0.05 as the frequency we correctly predict the mutation to be benign. We are 95% confident the prediction would lie between Benign and Likely benign for a frequency of 0.05.

Suggestions

To improve on this model we recommend expanding the dataset. Due to storage constraints, we were only able to work with chromosome 8 of the gnomAD dataset. Increasing this to the full exome will provide more consistent results across variables. We also recommend switching from a simple linear model since the prediction scores are categorical. An approach such as a logistic regression may be more applicable for this data. Logistic regression is used when the dependent variable is binary (e.g., yes/no, success/failure or in the context of our data benign vs not benign). It models the probability that the dependent variable belongs to a particular category based on one or more independent variables. Whereas, linear regression is used when modeling continuous numerical variables.

Logistic Regression uses the logistic function (also known as the sigmoid function) to transform the linear combination of predictor variables into probabilities bounded between 0 and 1. The logistic function converts the linear combination of predictor variables into log-odds, which are then transformed into probabilities. This allows logistic regression to provide estimates of the likelihood of an event occurring, such as the probability of pathogenicity given certain conditions such as allele frequency.